The term humoral is derived from the Latin word humor meaning "body fluid". Thus humoral immunity refers to immunity that can be conferred on a non-immune individual by administration of serum antibodies from an immune individual. Humoral immunity is mediated by molecules in blood and mucosal secretions containing antibodies that are produced by cells called B lymphocytes.

Antibodies recognize microbial antigens, neutralize infectivity of the microbes and target microbes for elimination by various effector mechanisms. Antibodies themselves are specialized and different types of antibodies may activate by different mechanisms, for example, IgG and IgM antibodies promote phagocytosis and IgE antibodies trigger the release of inflammatory mediators from leukocytes such as mast cells. Binding of antibody to antigen on a microorganism also can activate the complement system, resulting in lysis of the microbes.

Humoral immunity is the principal defense mechanism against extracellular microbes and their toxins because secreted antibodies can bind to these microbes and toxins and assist in their elimination.

10.3.2.2 Cell Mediated Immunity (CMI)

Cell Mediated immunity (cellular immunity) is mediated by T lymphocytes. There are two types of T-cells mainly TH and TC, which differ by their surface marker CD4 and CD8, respectively. T-cells are activated by antigen presenting cells (APC) after processing antigens. Both activated TH cells and TC cells serve as effector cells in CMI. Cytokines secreted by TH cells can activate various phagocytic cells, enabling them to phagocytose and kill microorganisms more effectively. Cytotoxic T lymphocytes (CTL) participate in CMI by killing altered self cells. They also play an important role in the killing of virus infected cells and tumor cells.

Intracellular microbes such as virus and some bacteria survive and proliferate inside phagocytes and other host cells, where they are inaccessible to circulating antibodies. Defense against such infections is a function of cell mediated immunity.

Antigen presenting cells are a functionally defined group of cells which are able to take up antigens and present them to T lymphocytes. Eg: Dendritic cells, activated macrophages, and activated B-cells.

A collective and coordinated response to the introduction of foreign substances in an individual, mediated by the cells and molecules of the immune system is referred to as immune response. An adaptive immune response that occurs upon the first exposure of native lymphocytes with a foreign antigen is known as primary immune response. An adaptive immune response that occurs upon the second or subsequent encounter of primed lymphocytes with a given antigen is known as secondary immune response.

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